GROUP 4: Project Proposal Week 1

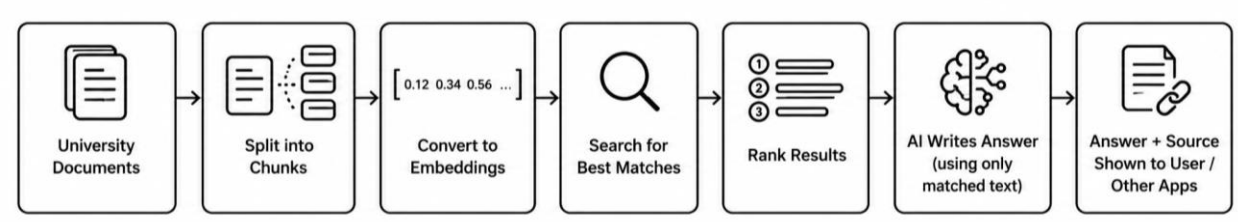
University Knowledge Assistant (RAG)

NAME: Aaron Jacob(J019), Vedant Kokate(J021),Saniya Lakdawala(J022), Sharvari Lokegaonkar(J023), Taha Lokhandwala(J024)

Problem	Students and staff waste a lot of time searching through scattered documents, rules, handbooks, exam guidelines, hostel rules, scholarship information, policies, calendars and circulars, just to find simple answers. Because there is no single place to look, people often ask friends or seniors instead, and get answers that are wrong or out of date.
Users	Students: for questions about exams, attendance, hostel, scholarships, etc. Faculty: for quick lookups of rules and processes. Admin staff: fewer repeated questions to answer. Other AI-University teams: can reuse this as a ready-made "search the rules" service instead of building their own (e.g. Group 10 Admin Copilot, Group 11 Communications).
Use Cases	• A student asks about attendance rules before an exam. • A faculty member checks a policy quickly while advising a student. • Another group's tool (like the Admin Copilot) calls our system instead of reading the raw PDF itself.
Input	A plain question typed in normal language, for example: "What is the minimum attendance required to appear for the end-semester exam?"
Output	A clear answer, plus proof of where it came from: • answer: the answer in simple language • source: which document and section it came from • confidence score: how sure the system is • supporting text: the exact lines used to answer
NLP Task	Retrieval-Augmented Generation (RAG) for university knowledge question answering. The system will: 1. Preprocess and split university documents into meaningful chunks. 2. Convert document chunks into embeddings. 3. Retrieve the most relevant chunks for a user's question. 4. Generate an answer using only the retrieved evidence. 5. Provide the supporting document and source section with the answer.
Dataset	The dataset will consist of official university documents such as academic regulations, examination guidelines, hostel rules, scholarship policies, academic calendars, and university circulars. Documents will be collected from official university sources and authorized internal platforms such as Teams, where permission is available. For evaluation, we will create approximately 60–100 sample question-answer pairs based on these documents. The questions will cover different

	university-related topics and will be divided into development/tuning and final testing sets.
Proposed Approach	Baseline (simple version): basic keyword search to find matching text, then show that text directly as the answer, no AI writing involved. Improved version: convert text into embeddings, search using meaning (not just keywords), and use an AI model to write a clear answer strictly based on the matched text, always showing the source. We will compare both versions on how accurate the search is, how correct the answers are, how often the system makes things up, and how correct the sources shown are. **Evaluation:** The baseline and improved approaches will be compared using retrieval accuracy, answer correctness, citation/source accuracy, and hallucination rate.
Institutional Value	Students and faculty get quick, trustworthy answers any time of day, with proof of source, reducing confusion caused by rumors or outdated information. Since this is built as a shared, reusable service, other groups' projects (like Admin Copilot or Communications) can plug into it instead of building their own search system from scratch.

Initial Architecture (How It Works)



What Already Exists / Literature & Product Review

Existing university FAQ and chatbot systems can answer common student questions, but many rely on predefined responses or keyword-based matching. RAG frameworks such as LangChain and LlamaIndex support document retrieval and AI-based question answering.

General AI assistants can also answer questions from uploaded documents, but they are not specifically designed as a reusable university knowledge service with institutional source citations.

Our project focuses on building a university-specific, reusable RAG asset that provides answers along with supporting documents and source sections, while evaluating retrieval accuracy, answer correctness, citation accuracy, and hallucination rate.